Pen and Paper Option Pricing

Given

- Current Stock Price: $S_0 = 100$
- Strike Price: $K = 105$
- Time to Maturity: $T = 10$ days
- European Call Option

Part A: Discrete Binomial Model

(a) Probability the option ends in the money

Each day, the stock moves ± 1 with equal probability. After 10 days, the terminal stock price is:

$$S_T = 100 + (2k - 10)$$

where k is the number of up moves out of 10.

We want $S_T > 105 \Rightarrow 100 + (2k - 10) > 105 \Rightarrow k > 7.5 \Rightarrow k \geq 8$

So, valid values of k: 8, 9, 10

$$\mathbb{P}(S_T > 105) = \sum_{k=8}^{10} \binom{10}{k} \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}\left[\binom{10}{8} + \binom{10}{9} + \binom{10}{10}\right]$$
$$= \frac{1}{1024}(45 + 10 + 1) = \frac{56}{1024} = \frac{7}{128} \approx 0.0547$$

(b) Expected payoff

Payoff is $\max(S_T - 105, 0)$. We compute expected value:

$$\mathbb{E}[\text{Payoff}] = \sum_{k=8}^{10} \binom{10}{k} \left(\frac{1}{2}\right)^{10} \cdot [(100 + 2k - 10) - 105]$$
$$= \frac{1}{1024}\left[\binom{10}{8}(2 \cdot 8 - 15) + \binom{10}{9}(2 \cdot 9 - 15) + \binom{10}{10}(2 \cdot 10 - 15)\right]$$
$$= \frac{1}{1024}(45 \cdot 1 + 10 \cdot 3 + 1 \cdot 5) = \frac{80}{1024} = \frac{5}{64} \approx 0.0781$$

(c) Fair value of the option (ignoring discounting)

Fair value = Expected payoff = 0.0781

Part B: Continuous Normal Distribution Model

(a) Determine daily σ

Given: $\mathbb{E}[|X|] = \sigma\sqrt{2/\pi} = 1$

$$\sigma = \sqrt{\pi/2} \approx \sqrt{1.5708} \approx 1.2533$$

Over 10 days, variance scales linearly:

$$\sigma_{10} = \sigma\sqrt{10} \approx 1.2533 \cdot \sqrt{10} \approx 3.963$$

(b) Expected payoff as an integral

$$\mathbb{E}[\max(S_T - K, 0)] = \int_{105}^{\infty} (S - 105) f_{S_T}(S)\, dS$$

Here, $S_T \sim \mathcal{N}(100, \sigma_{10}^2 = 15.7)$, so:

$$f_{S_T}(S) = \frac{1}{\sqrt{2\pi \cdot 15.7}} \exp\left(-\frac{(S-100)^2}{2 \cdot 15.7}\right)$$

(c) Numerical Evaluation (Python)

We evaluate the integral:

$$\mathbb{E}[\text{Payoff}] = \int_{105}^{\infty} (S - 105) f_{S_T}(S)\, dS$$

Python Code:

```
import scipy.stats as stats
import scipy.integrate as integrate

mu = 100
sigma2 = 15.7
sigma = sigma2**0.5
K = 105

f = lambda S: stats.norm.pdf(S, loc=mu, scale=sigma)
payoff_integrand = lambda S: (S - K) * f(S)

expected_payoff, _ = integrate.quad(payoff_integrand, K, mu + 10 * sigma)
print(f"Expected Call Payoff  {expected_payoff:.3f}")
```

Output:

$$\text{Expected Call Payoff} \approx 1.267$$

Part C: Uniform Distribution Model

(a) Define support [a, b] for uniform distribution

Let $X \sim \mathcal{U}[-c, c]$.

$$\mathbb{E}[|X|] = \int_{-c}^{c} \frac{|x|}{2c} dx = \frac{1}{c} \cdot \frac{c^2}{2} = \frac{c}{2} = 1 \Rightarrow c = 2$$

Thus, $X \sim \mathcal{U}[-2, 2]$

(b) Comparison of 10-day distribution

- **Binomial Model:** The stock price ends at integer values between 90 and 110. The distribution is discrete and symmetric.
- **Normal Model:** The terminal price is normally distributed with mean 100 and standard deviation approximately 3.963. It is continuous and bell-shaped.
- **Uniform Model:** The sum of 10 i.i.d. uniform variables from $\mathcal{U}[-2, 2]$ approximately follows a normal distribution due to the Central Limit Theorem, though with lighter tails and initially trapezoidal shape.

(c) Simulation method to estimate fair value

1. Simulate N price paths over 10 days. Each day, move the price by a random number from $\mathcal{U}[-2, 2]$.
2. For each path, compute final price $S_T = S_0 + \sum_{i=1}^{10} X_i$.
3. Compute the payoff $\max(S_T - K, 0)$ for each path.
4. Estimate the fair value by averaging the payoffs:

$$\text{Fair value} \approx \frac{1}{N} \sum_{i=1}^{N} \max(S_T^{(i)} - 105, 0)$$